

A short proof of the Observer Patch Holography simulation model

From a self-containing screen to Newton's constant

Paper release: r1466 Released: June 11, 2026

Selected-scale branch gravity theorem.

1. Purpose

The goal of this short note is to state the compact mathematical case that OPH is the correct fundamental theory of this Universe. The starting hypothesis is simple: reality is a simulation that contains itself. It contains observers, their records, and the rules by which those records are read back. Constants are therefore not external settings. They are fixed points of the closed readout structure.

Conventional particle-physics bookkeeping starts with the 19 fitted parameters of the minimal Standard Model, then adds gravitational and cosmological measured inputs such as G and Λ . OPH reduces the fundamental closure data to two dimensionless screen quantities. Their values are computable as unique fixed points of OPH readback laws. At the theorem target level, OPH is therefore a zero-parameter theory: numerical constants are readouts of the closed structure, with any remaining SI scale certificate stated as an explicit certificate target.

The first two OPH closure values are $P_\star$ and $N_\star$. $P_\star$ is the local pixel value of the screen. It says how much area and shared edge entropy one observer cell carries. $N_\star$ is the global capacity of the closed screen. It says how large the observer-facing de Sitter screen is. A finite screen simulator needs exactly these two dimensionless inputs: the size of one pixel and the capacity of the whole screen.

The selected scale branch supplies the no- G metrological coordinate $\gamma_\star$, equivalently $B_\star$. This is the bridge from the dimensionless OPH screen to SI metres. With $P_\star$, $N_\star$, and the selected scale certificate in place, the local area law has no adjustable Newton constant. The cell area and the shared edge entropy refer to the same observer cell, so $P_\star$ cancels and the area quantum becomes the Newton coupling.

This note proves the selected-branch gravity implication

$$(\bar{\ell}_{\text{shared}} = P_\star/4, \quad a_{\text{cell}} = P_\star \ell_\star^2, \quad \ell_\star^2 \text{ supplied by the scale certificate}) \implies G_{\text{SI}} = 6.674299995910528 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}.$$

It does not claim that $P_\star, N_\star$ alone determine an SI metre. The scale-freedom lemma in Appendix A proves that such a claim would be false. The scale certificate is stated explicitly.

Claim status. This is a selected-branch scale certificate. Once the no- G scale coordinate $\gamma_\star$, equivalently $B_\star$, is supplied, the OPH area coordinate and Newton constant are forced. A source-only proof of the SI value requires a source-only OPH readback map that emits $\gamma_\star$ or $B_\star$ without using measured G .

2. Fixed-point rule and the two closure variables

For any branch whose readback law is a contraction $F: I \rightarrow I$ on a complete metric interval,

$$F(I) \subseteq I, \quad d(F(x), F(y)) \leq q d(x, y), \quad 0 \leq q < 1,$$

Banach's theorem gives exactly one fixed point. Observation locates the branch interval; the contraction certificate removes numerical freedom inside that branch.

The compact OPH screen architecture has two dimensionless closure coordinates:

$$P \quad \text{local observer-cell pixel ratio,} \quad N \quad \text{global closed-screen capacity.}$$

P is local. It assigns one elementary observer cell its screen area and shared edge entropy:

$$a_{\text{cell}} = P \ell_\star^2, \quad \bar{\ell}_{\text{shared}} = P/4.$$

N is global. It assigns the closed observer screen its total capacity and therefore its dimensionless de Sitter branch:

$$N = \frac{3\pi}{\Lambda_\star \ell_\star^2}.$$

A finite OPH screen computation needs exactly these two closure coordinates before its readouts can have physical meaning: a pixel area/entropy unit for each cell, and a capacity for the whole closed screen.

Imported local pixel closure. On the declared OPH local-pixel branch, the readback equation

$$P = \Gamma(P) = \varphi + \frac{\sqrt{\pi}}{A_T(P)}$$

has a unique selected fixed point $P_\star$. This branch theorem is imported from the fine-structure surface. The gravity proof below does not require the numerical value of $P_\star$, because $P_\star$ cancels from the local area law. Informally, $P_\star$ is the one local pixel value for which geometric area and shared edge entropy describe the same observer cell.

Imported global capacity closure. On the declared OPH cosmic record-capacity branch, the readback equation

$$N = \mathcal{C}(N)$$

has a unique selected fixed point $N_\star$. Equivalently, on the count-density presentation,

$$N_\star = \text{MAR} \arg \max_N [\log |\Omega_N^{\text{sc}}| - N]$$

under the stated strict-concavity certificate. This branch theorem is imported from the capacity surface. Informally, $N_\star$ is the one total capacity for which the closed screen and its observer-facing de Sitter capacity name the same global object. A rough display is $N_\star \simeq 3.31 \times 10^{122}$.

Before Newton's constant is emitted, the capacity identity is written with the OPH area coordinate,

$$N_\star = \frac{A_{\text{dS}}}{4\ell_\star^2} = \frac{3\pi}{\Lambda_\star \ell_\star^2}.$$

The Planck-length notation is reserved for the downstream SI display after G_{SI} has been read out.

3. Dimensionless geometry from $P_\star, N_\star$

Let

$$P_\star = \frac{a_{\text{cell}}}{\ell_\star^2}, \quad N_\star = \frac{3\pi}{\Lambda_\star \ell_\star^2},$$

and define $B_\ell = \Lambda_\star N_\star$. Then

$$\Lambda_\star \ell_\star^2 = \frac{3\pi}{N_\star}, \quad \Lambda_\star a_{\text{cell}} = \frac{3\pi P_\star}{N_\star},$$

and

$$B_\ell \ell_\star^2 = 3\pi, \quad B_\ell a_{\text{cell}} = 3\pi P_\star.$$

These relations follow by substitution. They are exactly what the two dimensionless closure coordinates determine: dimensionless local/global geometry.

4. Scale certificate

$P_\star$ and $N_\star$ are dimensionless. They fix the local pixel ratio and the global screen capacity, then a branch coordinate must identify the SI length scale. OPH writes that coordinate as the no- G clock ratio $\gamma_\star$, equivalently $B_\star = \Lambda_\star N_\star$. The scale certificate has to come from an OPH readback map that does not use measured G , $\ell_P^2 = \hbar G/c^3$, or $\Lambda = 3\pi/(GN_\star)$. Appendix A contains the short scale-freedom proof.

5. Selected scale certificate

The primary scale coordinate is dimensionless:

$$\gamma_\star := \frac{\ell_\star \nu_{\text{Cs}}}{c},$$

where $\nu_{\text{Cs}} = 9\,192\,631\,770 \text{ s}^{-1}$ is the SI cesium frequency. Assume the selected OPH scale branch supplies the no- G clock ratio

$$\gamma_\star = 4.9559743365484194636657782433696431879484319705825 \times 10^{-34}.$$

Then

$$\ell_\star^2 = \left(\frac{\gamma_\star c}{\nu_{\text{Cs}}} \right)^2 = 2.61228030237427777887347769215451001461202676866 \times 10^{-70} \text{ m}^2.$$

Equivalently,

$$B_\star = \frac{3\pi}{\ell_\star^2} = 3.607873914680321576051841480160172547687708307217 \times 10^{70} \text{ m}^{-2}.$$

The displayed $B_\star$ is the SI form of the same scale certificate, not an additional pure OPH closure coordinate.

6. Local edge-entropy gravity law

For a cut with N_Σ cells,

$$A = N_\Sigma a_{\text{cell}}, \quad S_{\text{edge}} = N_\Sigma \bar{\ell}_{\text{shared}}.$$

If the area law is

$$S_{\text{edge}} = \frac{A}{4G_{\text{geom}}},$$

then substitution gives

$$N_\Sigma \bar{\ell}_{\text{shared}} = \frac{N_\Sigma a_{\text{cell}}}{4G_{\text{geom}}}, \quad G_{\text{geom}} = \frac{a_{\text{cell}}}{4\bar{\ell}_{\text{shared}}}.$$

7. Pixel cancellation

On the local OPH gravity branch,

$$a_{\text{cell}} = P_\star \ell_\star^2, \quad \bar{\ell}_{\text{shared}} = \frac{P_\star}{4}.$$

Therefore

$$G_{\text{geom}} = \frac{P_\star \ell_\star^2}{4(P_\star/4)} = \ell_\star^2.$$

The numerical value of G_{SI} is carried by the selected scale certificate $\gamma_\star$, equivalently $B_\star$. The local pixel $P_\star$ is load-bearing structurally: it makes the area and edge-entropy readings refer to the same local cell, and then cancels. The global capacity $N_\star$ is load-bearing in identifying the de Sitter scale branch through $B_\star = \Lambda_\star N_\star$. If $B_\star$ is supplied directly, $N_\star$ does not by itself determine the SI value.

8. SI Newton constant on the selected scale branch

By pixel cancellation,

$$G_{\text{geom}} = \ell_\star^2.$$

The SI display relation is

$$G_{\text{SI}} = \frac{c^3}{\hbar} G_{\text{geom}}.$$

Using

$$\ell_\star^2 = \left(\frac{\gamma_\star c}{\nu_{\text{Cs}}} \right)^2$$

gives

$$G_{\text{SI}} = \frac{c^5}{\hbar \nu_{\text{Cs}}^2} \gamma_\star^2.$$

With exact SI definitions,

$$c = 299792458 \text{ m s}^{-1}, \quad \nu_{\text{Cs}} = 9\,192\,631\,770 \text{ s}^{-1}, \quad \hbar = \frac{6.62607015 \times 10^{-34}}{2\pi} \text{ J s},$$

the selected-branch readout is

$$G_{\text{OPH}} = 6.674299995910528 \dots \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}.$$

CODATA reference:

$$G = 6.67430(15) \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}.$$

9. Derived readouts

Readout	Value	Meaning
$P_\star$	1.6309682094039593...	exact local pixel fixed point
$N_\star$	$\simeq 3.31 \times 10^{122}$	global closed-screen capacity fixed point
$\gamma_\star$	$4.95597433654841946 \dots \times 10^{-34}$	dimensionless selected scale certificate
$B_\star = \Lambda N_\star$	$3.60787391468 \dots \times 10^{70} \text{ m}^{-2}$	SI display of the scale certificate
$\ell_\star^2$	$2.61228030237 \dots \times 10^{-70} \text{ m}^2$	OPH area quantum
a_{cell}	$4.26054612722 \dots \times 10^{-70} \text{ m}^2$	physical area of one screen pixel
$\bar{\ell}_{\text{shared}}$	0.407742052350989825...	shared edge entropy per local cell
G_{geom}	$\ell_\star^2$	geometric Newton coupling
G_{SI}	$6.674299995910528 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$	SI Newton constant readout
Λ_{CRC}	$\frac{3\pi}{G_{\text{geom}} N_\star}$	downstream de Sitter display

10. TL;DR

OPH models reality as a closed self-referential mathematical structure. The observer does not invent the constants. The observer locates the experienced branch, and the OPH contraction laws return the unique fixed values on that branch.

$P_\star$ locks the local observer-cell pixel. $N_\star$ locks the total closed-screen capacity. These are the two quantities a screen computation must have: a pixel area/entropy unit and a total capacity. They determine the dimensionless local/global geometry. The selected scale certificate $\gamma_\star$, equivalently $B_\star$, locks the SI area quantum $\ell_\star^2$. Local edge entropy gives

$$G_{\text{geom}} = \frac{a_{\text{cell}}}{4\bar{\ell}_{\text{shared}}} = \ell_\star^2.$$

Exact SI definitions then give

$$G_{\text{SI}} = \frac{c^3}{\hbar} \ell_\star^2 = 6.674299995910528 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}.$$

If P is off its fixed point, the simulator has the wrong pixel. The local area reading and the local edge entropy reading stop describing one stable observer cell. If N is off its fixed point, the simulator has the wrong screen. The closed screen does not reproduce its own capacity branch. If the scale branch is off by r in length, the SI gravity readout is off by r^2 .

The proof contains no adjustable Newton constant. Gravity is the area readout of the same observer-cell closure that defines the local screen unit. The selected-scale statement is the gravitational selection test for OPH: the closed observer-patch model plus the no- G scale certificate produces the measured SI coupling. The agreement lands on the CODATA value because the scale certificate gives the area quantum first, and SI then names the same area as Newton's constant.

Appendix A. Audit checks

Scale freedom lemma. No expression using only $P_\star, N_\star$, pure dimensionless constants, and exact display constants $c, \hbar$ determines B_ℓ in m^{-2} , $\ell_\star^2$ in m^2 , or G_{SI} in SI units.

For any $\lambda > 0$, rescale

$$\ell_\star^2 \mapsto \lambda^2 \ell_\star^2, \quad a_{\text{cell}} \mapsto \lambda^2 a_{\text{cell}}, \quad \Lambda_\star \mapsto \lambda^{-2} \Lambda_\star.$$

Then both closure coordinates are unchanged:

$$\frac{\lambda^2 a_{\text{cell}}}{\lambda^2 \ell_\star^2} = \frac{a_{\text{cell}}}{\ell_\star^2}, \quad \frac{3\pi}{(\lambda^{-2} \Lambda_\star)(\lambda^2 \ell_\star^2)} = \frac{3\pi}{\Lambda_\star \ell_\star^2}.$$

The same $P_\star, N_\star$ therefore admit

$$B_\ell \mapsto \lambda^{-2} B_\ell, \quad G_{\text{SI}} \mapsto \lambda^2 G_{\text{SI}}.$$

A separate scale certificate is required for the SI gravity row.

Dependency ledger.

Object	Allowed role	Forbidden upstream use
$P_\star$	local pixel fixed point	does not set metres by itself
$N_\star$	global screen-capacity fixed point	does not set metres by itself
$\gamma_\star$	dimensionless no- G scale certificate	must not be fitted from measured G
$B_\star$	SI display of the scale certificate	must not be computed from $3\pi/(GN_\star)$
$\ell_\star^2$	area coordinate from $3\pi/B_\star$	must not be computed from $\hbar G/c^3$
G_{geom}	local edge-entropy output	not an input to the scale certificate
Λ_{CRC}	downstream de Sitter display	evaluated only after G_{geom}

Minimal metrological bridge. Fix the exact SI constants c , $\hbar$, and the cesium clock frequency $\nu_{\text{Cs}} = 9\,192\,631\,770\text{ s}^{-1}$. A numerical SI Newton constant is equivalent to the dimensionless scale ratio

$$\gamma_G := \left(\frac{\hbar G_{\text{SI}} \nu_{\text{Cs}}^2}{c^5} \right)^{1/2}.$$

The units cancel:

$$\frac{(\text{kg m}^2 \text{s}^{-1})(\text{m}^3 \text{kg}^{-1} \text{s}^{-2})(\text{s}^{-2})}{(\text{m s}^{-1})^5} = 1.$$

Solving the definition gives

$$G_{\text{SI}} = \frac{c^5}{\hbar \nu_{\text{Cs}}^2} \gamma_G^2.$$

Thus a no- G OPH emission of $\gamma_\star$ is exactly a no- G SI scale certificate.

Equivalent scale charts. Given $N_\star$, any one of

$$\ell_\star^2, \quad B_\star = \Lambda_\star N_\star, \quad \Lambda_\star, \quad R_\star = \sqrt{3/\Lambda_\star}, \quad t_\star = R_\star/c, \quad \gamma_\star = \ell_\star \nu_{\text{Cs}}/c$$

determines all the others. The conversion formulas are

$$B_\star = \frac{3\pi}{\ell_\star^2}, \quad \Lambda_\star = \frac{B_\star}{N_\star}, \quad R_\star = \sqrt{\frac{3N_\star}{B_\star}},$$

$$t_\star = \frac{1}{c} \sqrt{\frac{3N_\star}{B_\star}}, \quad \gamma_\star = \frac{\nu_{\text{Cs}}}{c} \sqrt{\frac{3\pi}{B_\star}}.$$

These are chart changes for one scale certificate, not independent constants.

Anti-smuggling criterion. Let $\mathcal{B}$ be the scale certificate used to compute G_{SI} . The numerical gravity row is a no- G derivation only if the dependency graph for $\mathcal{B}$ contains no path from

$$G_{\text{exp}}, \quad \ell_P^2 = \frac{\hbar G}{c^3}, \quad \Lambda = \frac{3\pi}{GN_\star}, \quad B_\ell = \frac{3\pi c^3}{\hbar G}$$

to $\mathcal{B}$. If such a path exists, the row is a calibration of G . If no such path exists, the algebraic computation of G_{SI} from $\mathcal{B}$ contains no measured- G dependency.

Pixel insensitivity. On the local OPH gravity branch,

$$a_{\text{cell}} = P\ell_\star^2, \quad \bar{\ell}_{\text{shared}} = P/4, \quad G_{\text{geom}} = \frac{a_{\text{cell}}}{4\bar{\ell}_{\text{shared}}}.$$

Substitution gives

$$G_{\text{geom}} = \frac{P\ell_\star^2}{4(P/4)} = \ell_\star^2, \quad \frac{\partial G_{\text{geom}}}{\partial P} = 0.$$

The local pixel value is structurally necessary because it identifies the two readings of one cell. The numerical area readout is insensitive to the coordinate P after that identification.

Planck area is downstream. If the OPH local gravity row emits

$$G_{\text{SI}} = \frac{c^3}{\hbar} \ell_\star^2,$$

then the SI Planck area computed after the emission is

$$\ell_{P,\text{OPH}}^2 := \frac{\hbar G_{\text{SI}}}{c^3} = \ell_\star^2.$$

The symbol ℓ_P^2 can therefore name the emitted area downstream. It cannot be used upstream to derive it.

Coupled closure uniqueness target. Let $K = I_N \times I_\gamma$ be a complete product metric space with metrics d_N, d_γ . Let

$$\mathcal{S}(N, \gamma) = (S_N(N, \gamma), S_\gamma(N, \gamma))$$

be the joint capacity/scale certificate map. Suppose there is a nonnegative matrix

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \rho(M) < 1,$$

such that

$$\begin{aligned} d_N(S_N(x), S_N(y)) &\leq a d_N(N_x, N_y) + b d_\gamma(\gamma_x, \gamma_y), \\ d_\gamma(S_\gamma(x), S_\gamma(y)) &\leq c d_N(N_x, N_y) + d d_\gamma(\gamma_x, \gamma_y). \end{aligned}$$

Perron–Frobenius gives a positive vector $w = (w_N, w_\gamma)$ and $q < 1$ with $Mw \leq qw$. The weighted metric

$$d_w(x, y) = \max \left\{ \frac{d_N(N_x, N_y)}{w_N}, \frac{d_\gamma(\gamma_x, \gamma_y)}{w_\gamma} \right\}$$

makes $\mathcal{S}$ a contraction. Banach’s theorem gives a unique joint fixed point $(N_\star, \gamma_\star)$ on K .

Residual and sensitivity certificates. If $R : I \rightarrow I$ is a contraction with constant $q < 1$, and a candidate x_0 satisfies $d(R(x_0), x_0) \leq \varepsilon$, then the fixed point $x_\star$ obeys

$$d(x_\star, x_0) \leq \frac{\varepsilon}{1 - q}.$$

For equivalent scale coordinates,

$$\frac{\delta G}{G} = \frac{\delta \ell_\star^2}{\ell_\star^2} = 2 \frac{\delta \gamma_\star}{\gamma_\star} = - \frac{\delta B_\star}{B_\star}.$$

These two formulas convert a numerical readback residual into a certified interval and show exactly how scale-coordinate errors move the gravity row.

Frozen-branch retuning test. Let $\mathcal{B}_\star$ denote the published scale certificate, written as $B_\star$, $\ell_\star^2$, or $\gamma_\star$. If $\mathcal{B}_\star$ is changed after a new measurement of G is available, and the change is not forced by a pre-existing no- G OPH readback certificate, then the numerical gravity row is a calibration rather than a prediction.

Appendix B. OPH-predicted values

The short proof isolates the gravity row. The wider OPH claim is that the same closure architecture fixes the local pixel, total screen capacity, support-visible spacetime branch, realized gauge branch, and declared quantitative particle branches. The table separates direct measurements, public limits, definitions, and values whose public measurement does not exist.

A numerical σ is shown only when the cited public source gives a central value and a one-standard-deviation uncertainty. Exact definitions and rounded matches are marked as exact hits. Limit rows, structural rows, and non-Gaussian global-fit profiles are marked by status rather than forced into a fake scalar sigma. Rows whose OPH value carries more digits than public measurement permits are sharper OPH readouts if the branch derivation is accepted.

Lane	OPH prediction	Measurement or public status	Residual	Source
Local pixel	$P_\star = 1.6309682094039593 \dots$	No direct laboratory measurement. This is the local cell-area and edge-entropy fixed point required by a finite OPH screen.	n/a	OPH
Local cell entropy	$\bar{\ell}_{\text{shared}} = P_\star / 4 = 0.407742052350989825 \dots$	No direct laboratory measurement. It is the entropy weight assigned to one observer cell by the same fixed point that fixes $P_\star$.	n/a	OPH
Global capacity	$N_\star \simeq 3.31 \times 10^{122}$	No direct count measurement. Cosmological comparison uses de Sitter capacity after the scale branch is supplied.	n/a	OPH
Scale certificate	$\gamma_\star = 4.95597433655 \dots \times 10^{-34}$ $B_\star = 3.60787391468 \dots \times 10^{70} \text{m}^{-2}$ $\ell_\star^2 = 2.61228030237 \dots \times 10^{-70} \text{m}^2$	No direct public measurement. The gravity row below is the scalar test of this selected SI scale certificate.	n/a	OPH
Area quantum and pixel area	$a_{\text{cell}} = 4.26054612722 \dots \times 10^{-70} \text{m}^2$ $G_{\text{geom}} = \ell_\star^2$	No direct public measurement of an OPH screen cell. The inferred area quantum is tested through G_{SI} .	n/a	OPH
Newton coupling	$G_{\text{SI}} = 6.674299995910528 \times 10^{-11}$	$G = 6.67430(15) \times 10^{-11} \text{m}^3 \text{kg}^{-1} \text{s}^{-2}$. OPH prints more digits than the CODATA uncertainty permits.	$-2.7 \times 10^{-5} \sigma$	S1
Cosmological constant	$\Lambda_{\text{CRC}} \approx 1.09 \times 10^{-52} \text{m}^{-2}$	Planck 2018 gives the late-universe Λ CDM parameter surface; the rounded OPH value agrees at displayed order.	exact hit at shown precision	S5
Static-patch rate	$H_{\text{dS}} \approx 55.759940256 \text{ km s}^{-1} \text{ Mpc}^{-1}$	This is a de Sitter static-patch display, not the fitted local Hubble constant.	not a fitted H_0	OPH
Hubble benchmark	$H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ in compressed diagnostics	Planck 2018 reports $H_0 = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$. This row is a benchmark input in the compressed comparison.	0σ	S5
Lorentz group	$\text{Conf}^+(S^2) \cong \text{SO}^+(3, 1)$	Structural prediction. Direct scalar sigma is not the right statistic; the comparison is the Lorentz-kinematics branch.	n/a	OPH
Causal speed	$c = 299792458 \text{ m s}^{-1}$	Exact SI definition. This is a display convention for the Lorentz speed.	exact hit by definition	S3
Einstein branch	Jacobson-type local Einstein equation	Structural prediction. Tests are GR/weak-field/null-energy comparisons rather than one scalar value.	n/a	OPH
Gauge quotient	$\text{SU}(3)_c \times \text{SU}(2)_L \times \text{U}(1)_Y / \mathbb{Z}_6$	Matches the observed Standard Model gauge quotient. No one-number sigma.	exact structural hit	S6
Hypercharge, color, generations	one-generation hypercharge lattice $N_c = 3, \quad N_g = 3$	Matches observed Standard Model charge and family structure. No one-number sigma.	exact structural hit	S6

Lane	OPH prediction	Measurement or public status	Residual	Source
Massless protected carriers	$m_\gamma = 0$	Photon and gluon masslessness are gauge-branch structural hits. Graviton masslessness belongs to the diffeomorphism/Lorentz gravity branch. Direct measurements are public upper limits rather than exact mass measurements.	γ : structural hit	S6
	$m_g = 0$		g : structural hit	
	$m_{\text{grav}} = 0$		grav : structural hit	
Gauge proton decay	$\tau_p^{(\text{gauge})} = \infty$	Public searches give lower limits on proton decay channels. The OPH gauge row predicts no simple-GUT gauge boson decay mode.	n/a	S6
Fine-structure source trunk	$\alpha_{\text{source}}^{-1} = 136.994835164621649\dots$	No direct public measurement. It is the source calculation before the declared hadronic/same-scheme endpoint payload.	n/a	OPH
Fine-structure endpoint	$\alpha^{-1}(0) = 137.035999177(21)$ $\alpha(0) = 0.00729735256433\dots$	CODATA gives $\alpha^{-1} = 137.035999177(21)$. The public endpoint is not used in the gravity proof.	0σ	S2
Electroweak source	$\alpha_2(m_Z) = 0.03377843630219015$ $\alpha_Y(m_Z) = 0.010131601067241624$	Electroweak-fit comparison row. A scalar sigma depends on scheme and fit covariances, so it is not quoted here.	n/a	S6
Electroweak scale	$v = 246.76711732749683 \text{ GeV}$	Derived from the electroweak branch and displayed in the particle surface. Direct comparison depends on G_F convention and loop scheme.	n/a	S6
W/Z validation pair	$M_W = 80.377 \text{ GeV}$ $M_Z = 91.18797809193725 \text{ GeV}$	PDG 2024 gives $M_W = 80.3692 \pm 0.0133 \text{ GeV}$ and $M_Z = 91.1876 \pm 0.0021 \text{ GeV}$.	0.59σ 0.18σ	S6
Higgs/top branch	$m_H = 125.1995304097179 \text{ GeV}$ $m_t = 172.35235532883115 \text{ GeV}$ $m_e = 0.00051099895$	PDG 2024 gives $m_H = 125.20 \pm 0.11 \text{ GeV}$ and direct $m_t = 172.57 \pm 0.29 \text{ GeV}$.	-0.004σ -0.75σ	S6,S7
Charged leptons	$m_\mu = 0.1056583755 \text{ GeV}$ $m_\tau = 1.7769324651340912$ $u = 0.00216, \quad d = 0.00470$ $s = 0.0935, \quad c = 1.273$	PDG lepton summary gives the displayed central values. The tau residual is 0.03σ ; e, μ match at displayed precision.	$\leq 0.03\sigma$	S8
Quark sextet	$b = 4.183$ $t = 172.35235532883115 \text{ GeV}$ 0.017454720257976796	PDG quark summary gives the displayed light/heavy quark central values. The top comparison is listed in the Higgs/top row.	0σ at shown precision	S7
Neutrino masses	$0.019481987935919015 \text{ eV}$ 0.05307522145074924	Individual absolute neutrino masses are not yet measured. KATRIN gives a direct upper limit on the effective electron-neutrino mass.	n/a	S10
Neutrino sum	$\sum m_\nu = 0.09001192964464505 \text{ eV}$	Planck 2018 plus BAO gives $\sum m_\nu < 0.12 \text{ eV}$ at 95% CL. OPH lies below the public upper limit.	below limit	S5,S11
Neutrino mixing display	$\theta_{12} = 34.2259^\circ, \theta_{23} = 49.7228^\circ$ $\theta_{13} = 8.68636^\circ, \delta = 305.581^\circ$	NuFIT 6.0 gives global-fit profiles. The profiles are non-Gaussian and ordering-dependent, so a single scalar sigma is not quoted.	n/a	S9
Majorana phases	$\alpha_{21} = 153.618518^\circ$ $\alpha_{31} = 257.003241^\circ$	Majorana phases are not yet measured. They require neutrinoless double-beta or equivalent absolute-phase information.	n/a	S11
Compressed matter budget	$\Omega_{\Lambda, \text{OPH}} = 0.684423332$ $\Omega_A = 0.264114401$ $\Omega_m = 0.315484968$	Planck 2018 gives $\Omega_m = 0.315 \pm 0.007$. Flat Λ CDM gives $\Omega_\Lambda \simeq 0.685 \pm 0.007$.	$\Omega_m : 0.07\sigma$ $\Omega_\Lambda : -0.08\sigma$	S5

Measurement source keys.

- S1: NIST CODATA 2022 G .
S2: NIST CODATA 2022 α^{-1} .
S3: NIST CODATA c , exact.
S4: NIST CODATA $\hbar$, exact. Used in the arithmetic certificate.
S5: Planck 2018 cosmological parameters.
S6: PDG 2024 gauge and Higgs summary.
S7: PDG 2024 quark summary.
S8: PDG 2024 lepton summary.
S9: NuFIT 6.0 oscillation fit.
S10: KATRIN 2025 direct neutrino-mass limit.
S11: PDG 2024 neutrino-mass review.

OPH paper keys.

- O1: Observers Are All You Need.
O2: Reality as Consensus Protocol.
O3: Screen Microphysics and Observer Synchronization.
O4: Recovering Relativity and Standard Model Structure from Observer Overlap Consistency.
O5: Deriving the Particle Zoo from Observer Consistency.